Bivariate Normal Distribution

1 Definition

A bivariate RV (X, Y) is said to have a bivariate normal density if its PDF is given by,

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2+\left(\frac{y-\mu_2}{\sigma_2}\right)^2-2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right)\right\}}$$

where, $\mu_1, \mu_2 \in \mathcal{R}, \sigma_1, \sigma_2 > 0, |\rho| < 1$.
Notation: $(X, Y) \sim \text{BVN}\left(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho\right)$

$$(X, Y) \sim N_2\left(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho\right)$$

2 Marginal Distribution

The exponent of BVN is,

$$\begin{aligned} Q(x,y) &= \frac{1}{1-\rho^2}\left\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(\frac{y-\mu_2}{\sigma_2}\right)^2\right\} \\ &= \frac{1}{1-\rho^2}\left[\left\{\left(\frac{y-\mu_2}{\sigma_2}\right) - \rho\left(\frac{x-\mu_1}{\sigma_1}\right)\right\}^2 + \left\{\left(1-\rho^2\right)\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right\}\right] \\ &= \frac{\left\{y-\mu_2-\rho\frac{\sigma_2}{\sigma_1}(x-\mu_1)\right\}^2}{\sigma_2^2(1-\rho^2)} + \left(\frac{x-\mu_1}{\sigma_1}\right)^2 \\ &= \frac{\{y-\eta(x)\}^2}{\sigma_{2.1}^2} + \left(\frac{x-\mu_1}{\sigma_1}\right)^2 \end{aligned}$$

$$\eta(x) = \mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x-\mu_1)$$
$$\sigma_{2.1}^2 = \sigma_2^2\left(1-\rho^2\right)$$
$$\Rightarrow \sigma_{2.1} = \sigma_2\sqrt{1-\rho^2}$$

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{XY}(x,y)dy \quad x \in \mathcal{R} \\ &= \frac{e^{-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2}}{\sigma_1\sqrt{2\pi}} \int_{-\infty}^{\infty} \underbrace{\frac{e^{-\frac{1}{2}\left\{\frac{y-\eta(x)}{\sigma_{2.1}}\right\}^2}}{\sqrt{2\pi}\sigma_{2.1}}}_{\text{pdf of } N\left(\eta(x),\sigma_{2.1}^2\right)} dy \\ &= \frac{1}{\sigma_1\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-u_1}{\sigma_1}\right)^2}, \quad x \in \mathbb{R}. \end{aligned}$$

Hence. $X \sim N\left(\mu_1, \sigma_1^2\right)$
Similarly $Y \sim N\left(\mu_2, \sigma_2^2\right)$

3 Conditional Distribution

The conditional PDF of $Y \mid K = x$ is.

$$f_{Y|X}(y \mid x) = \frac{f_{XY}(x,y)}{f_X(x)}, \quad y \in \mathcal{R}$$

$$= \frac{\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2}\left\{\frac{(y-\eta(x))^2}{\sigma_{2\cdot1}^2}+\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right\}}}{\frac{1}{\sigma_1\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2}}$$

Hence, $Y \mid X = x \sim N\left(\eta(x), \sigma_{2.1}^2\right)$
Note that, $E(Y \mid X = x) = \eta(x) = \mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x - \mu_1)$
$\Rightarrow$ The regression equation of y on x is linear and $\text{Var}(Y \mid X = x) = \sigma_{2.1}^2 = \sigma_2^2\left(1 - \rho^2\right), x \in \mathbb{R}$.
$\Rightarrow$ The conditional distribution of Y given $X = x$ is homocadastic.

Remark:

(1) $(X, Y) \sim \text{BVN}\left(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho\right)$

$$\Rightarrow X \backsim N\left(\mu_1, \sigma_1^2\right). \quad \Rightarrow \mu_1 = E(X), \quad \sigma_1^2 = V(X)$$
$$\Rightarrow Y \sim N\left(\mu_2, \sigma_2^2\right). \quad \Rightarrow \mu_2 = E(Y), \quad \sigma_2^2 = V(Y)$$

Note,

$$\begin{aligned} E(XY) &= E\{E\{XY \mid X)\} \\ &= E\{XE(Y \mid X)\} \\ &= E\left\{X \quad \left\{\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x - \mu_1)\right\}\right\} \\ &= \mu_1\mu_2 + \rho\frac{\sigma_2}{\sigma_1}E\left(X\left(X - \mu_1\right)\right) \\ &= \mu_1\mu_2 + \rho\frac{\sigma_2}{\sigma_1}\sigma_1^2 \quad E(X(X-\mu)) = E[(X-\mu)(X-\mu)] \end{aligned}$$

$$E(XY) - \mu_1\mu_2 = \rho\sigma_2\sigma_1$$
$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{V(X)}\sqrt{V(Y)}} = \frac{\rho\sigma_2\sigma_1}{\sigma_1\sigma_2} = \rho$$

For a bivariate $RV\binom{X}{Y}$, the dispersion matrix (variance-covariance matrix) is given by,
$\begin{pmatrix} \text{cov}(X,X) & \text{cov}(X,Y) \\ \text{cov}(X,Y) & \text{cov}(Y,Y) \end{pmatrix} = \begin{pmatrix} V(X) & \text{cov}(X,Y) \\ \text{cov}(X,Y) & V(Y) \end{pmatrix} = \sum^{2\times2}$
This is a symmetric non-negative matrix (nnd).
Note that, $|\Sigma| = \begin{vmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{vmatrix} = \sigma_1^2\sigma_2^2\left(1 - \rho^2\right)$. Clearly, $|\Sigma| \geqslant 0$.
If $|\Sigma| = 0$ ie. Σ is singular, the distribution of (X, Y) is called a singular distribution in $\mathcal{R}^2$.
If $|\Sigma| > 0$ ie. Σ is non-singular, then the distribution is called a non-singular distribution in $\mathcal{R}^2$.
If $\rho = \pm1$, then $|\Sigma| = 0$ ie. the distribution is singular in $\mathbb{R}^2$

$$p[aX + bY = c] = 1 \text{ for some } (a, b) \neq (0, 0)$$

If $|\rho| < 1$, the distribution in $\mathbb{R}^2$ is non-singular.
If $|\rho| = 1$, the distribution has no PDF in $\mathbb{R}^2$.

1. Denote $\phi(x, y, \rho) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{x^2+y^2-2\rho xy}{2(1-\rho^2)}}, (x, y) \in \mathbb{R}^2$ is a PDF of $BVN(0, 0, 1, 1, \rho)$. Define, $f_{X,Y}(x, y) = \frac{1}{2}\{\phi(x, y, \rho) + \phi(x, y, -\rho)\}$. Find the marginal distribution of X, Y. Also find ρ_{XY} and comment.

The marginal distribution of $\phi(x, y, \rho)$ are both $N(0, 1)$ with correlation coefficient ρ.

Now,

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{XY}(x,y)dy, \quad x \in \mathbb{R} \\ &= \frac{1}{2}\left\{\int_{-\infty}^{\infty} \phi(x,y,\rho)dy + \int_{-\infty}^{\infty} \phi(x,y,-\rho)dy\right\} \\ &= \frac{1}{2}\{\phi(x)+\phi(x)\} = \phi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}, x \in \mathbb{R} \end{aligned}$$

Similarly, $f_Y(y) = \phi(y) = \frac{1}{\sqrt{2\pi}}e^{-y^2/2}, y \in \mathbb{R}$.

Note that, the marginal distribution of X and Y are both $N(0,1)$. But the joint PDF is not BVN PDF. Hence, from the knowledge of marginal distribution the joint distribution can't be determined uniquely.

$$\begin{aligned} \rho(X,Y) &= E(XY) \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy f_{XY}(x,y)dxdy \\ &= \frac{1}{2}\left\{\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\phi(x,y,\rho)dxdy + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\phi(x,y,-\rho)dxdy\right\} \\ &= \frac{1}{2}\{\rho+(-\rho)\} = 0 \end{aligned}$$

But $f_{XY}(x,y) \neq f_X(x)f_Y(y)$. Hence, $\rho_{XY} = 0 \nRightarrow X, Y$ are independent.

Notation $\binom{X}{Y} \sim BN\left(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho\right)$.
$\binom{X}{Y} \sim N_2\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}\right)$

2. Consider a bivariate PDF of (X,Y),

$$f_{XY}(x,y) = \frac{1}{2\pi}e^{-\frac{x^2+y^2}{2}\left\{1+xye^{-\frac{x^2+y^2}{2}}\right\}}, \quad (x,y) \in R^2$$

Find the PDFS of X, Y. Hence, comment- From knowledge of marginal distribution, the joint dist can't be determined uniquely.

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{XY}(x,y)dy, \quad x \in R. \\ &= \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-y^2/2}dy + \frac{1}{2\pi}xe^{-x^2}\int_{-\infty}^{\infty}\underbrace{ye^{-y^2}}_{\text{odd function}}dy \\ &= \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}} = \phi(x) \end{aligned}$$

Here, $X \sim N(0,1)$. Similarly $Y \backsim N(0,1)$.

Also, if $(X,Y) \backsim \text{BN}(0,0,1,1,p)$, we also have $X \sim N(0,1), Y \sim N(0,1)$. But, if $X, Y \sim N(0,1)$, the joint dist may be $BN(0,0,1,1,p)$ or may be given by $f_{X,Y}(x,y)$. Hence, the statement given is correct.

4 Theorem

Let $(X,Y) \sim BN$ with correlation coefficient ρ. Then (X,Y) are independently distribution iff $\rho = 0$.

5 Moment Generating Function

$$(X,Y) \sim BN\left(\mu_1, \mu_2, \sigma^2, \sigma_2^2, \rho\right)$$

$$\text{Define,} \quad U = \frac{X-\mu_1}{\sigma_1}, \quad V = \frac{Y-u_2}{\sigma_2}$$

$(U,V) \sim \text{BVN}(0,0,1,1,\rho)$.

$$\begin{aligned}
M_{U,V}(t_1,t_2) =& E\left[e^{t_1U+t_2V}\right] = E\left[E\left[e^{t_1U+t_2V} \mid U\right]\right] \\
=& E\left[e^{t_1U}E\left[e^{t_2v} \mid U\right]\right] (\longrightarrow MGF \text{ of V/U }) \\
&= E\left[e^{t_1U}M_{V|U}(t_2)\right] \\
&= E\left[e^{t_1U}e^{\rho u t_2+\frac{1}{2}t_2^2\left(1-\rho^2\right)}\right] \\
&= e^{\frac{1}{2}t_2^2\left(1-\rho^2\right)}E\left[e^{(t_1+\rho t_2)U}\right] \\
&= e^{\frac{1}{2}t_2^2\left(1-\rho^2\right)}M_U(t_1+\rho t_2) \\
&= e^{\frac{1}{2}t_2^2\left(1-\rho^2\right)}e^{\frac{1}{2}(t_1+\rho t_2)^2} \\
&= e^{\frac{1}{2}\left(t_1^2+t_2^2+2\rho t_1t_2\right)}
\end{aligned}$$

$$(U,V) \sim BN(0,0,1,1,\rho) \quad \Rightarrow U \sim N(0,1), V/U = u \sim N\left(\rho u, 1-\rho^2\right)$$

MGF of (X,Y),

$$\begin{aligned}
\mathrm{M}_{XY}(t_1,t_2) &= E\left[e^{t_1X+t_2Y}\right] \\
&= E\left[e^{t_1(\mu_1+\sigma_1U)+t_2(\mu_2+\sigma_2V)}\right] \\
&= e^{t_1\mu_1+t_2\mu_2}E\left[e^{t_1\sigma_1U+t_2\sigma_2V}\right] \\
&= e^{t_1\mu_1+t_2\mu_2}\,\mathrm{M}_{UV}(t_1\sigma_1,t_2\sigma_2) \\
&= e^{t_1\mu_1+t_2\mu_2+\frac{1}{2}\left\{t_1^2\sigma_1^2+t_2^2\sigma_2^2+2\rho t_1t_2\sigma_1\sigma_2\right\}}
\end{aligned}$$

6 Theorem

Let, $(X,Y) \sim$ BVN with correlation coefficient ρ. Then, X and Y are independently distributed iff $\rho = 0$.

If part,

$$\begin{aligned}
\rho = 0. \Rightarrow f_{XY}(x,y) &= \frac{1}{2\pi\sigma_1\sigma_2}e^{-\frac{1}{2}\left(\frac{x-u_1}{\sigma_1}\right)^2+\left(\frac{y-u_2}{\sigma_2}\right)^2} \\
&= f_X(x)f_Y(y), \quad (x,y) \in \mathcal{R}^2
\end{aligned}$$

7 Problem

a) Let, $(X,Y) \sim BN\left(\mu_x,\mu_y,\sigma_x^2,\sigma_y^2,\rho\right)$. Then, find the distribution of $Z = aX+bY+c$.

$$\begin{aligned}
M_Z(t) &= E\left[e^{t(aX+bY+c)}\right] = e^{tc}M_{XY}(ta,tb) \\
&= e^{tc}\left[e^{ta\mu_1}+tb\mu_2+\frac{1}{2}\left\{t^2a^2\sigma_1^2+t^2b^2\sigma_2^2+2\rho t^2ab\sigma_1\sigma_2\right\}\right. \\
&= e^{t(a\mu_1+b\mu_2+c)+\frac{1}{2}t^2\left(a^2\sigma_1^2+b^2\sigma_2^2+2\rho ab\sigma_1\sigma_2\right)}
\end{aligned}$$

By uniqueness of MGF we can say, $Z \sim N\left(a\mu_1+b\mu_2+c, \sigma^2a_1^2+b^2\sigma_2^2+2\rho ab\sigma_1\sigma_2\right)$

b) Let, $(X,Y) \sim \mathrm{BN}(0,0,1,1,\rho)$. Find the constant a,b,c,d such that $U = aX+bY$, $v = cX+dY$ are independent.

If $(X,Y) \sim BVN\left(\mu_X,\mu_Y,\sigma_X^2\sigma_Y^2,\rho\right)$.
Two any inear combination follows univariate Normal. They may have different parameters.

$$\begin{aligned}
M_{U,V}(t_1,t_2) &= E\left[e^{t_1U+t_2V}\right] = E\left[e^{t_1(aX+bY)+t_2(cX+dY)}\right] \\
&= E\left[e^{(t_1a+t_2c)X+(t_1b+t_2d)Y}\right]
\end{aligned}$$

$$= M_{X,Y}\left(t_1a + t_2c, t_1b + t_2d\right)$$
$$= e^{\frac{1}{2}\left\{(t_1a+t_2b)^2+(t_1b+t_2d)^2+2\rho(t_1a+t_2c)(t_1b+t_2d)\right\}}$$
$$= e^{\frac{1}{2}\left\{t_1^2\left(a^2+b^2+2ab\rho\right)+t_2^2\left(c^2+d^2+2\rho cd\right)+2t_1t_2(ac+bd+\rho(ad+cb))\right\}}$$

which is again a MGF of BVN with $\text{cov}(U,V) = ac + bd + \rho(ad + bc)$.
$\Rightarrow U, V$ are independently distributed if $\text{Cov}(U,V) = 0$ iff $ac + bd + \rho(ad + cb) = 0$.

c) Let, $X_1, X_2, \ldots, X_n$ be a random sample from $N(\mu, \sigma_2)$. Find the joint distribution of $U = \sum_{i=1}^n a_iX_i = a'x$ and $V = \sum_{i=1}^n b_iX_i = b'x$ i) when and only when we u and v independently distribution? ii) Show that, $X_1 - \bar{X}, X_2 - \bar{X}, \ldots, X_n - \bar{X}$ are independently distributed. Hence, show that $\bar{X}$ and s^2 are independently distributed.

$$M_{UV}(t_1,t_2) = E\left[e^{t_1\sum a_iX_i + t_2\sum b_iX_i}\right]$$
$$= E\left[e^{\sum_{i=1}^n (t_1a_i+t_2b_i)X_i}\right] = \prod_{i=1}^n M_{X_i}\left(t_1a_i + t_2b_i\right)$$
$$= \prod_{i=1}^n \left(e^{(t_1a_i+t_2b_i)\mu+\frac{1}{2}(t_1a_i+t_2b_i)^2\sigma^2}\right)$$
$$= \exp\left[t_1\left(\sum a_i\right)\mu + t_2\left(\sum b_i\right)\mu + \frac{1}{2}\left\{t_1^2\left(\sum a_i^2\right)\sigma^2 + t_2^2\left(\sum b_i^2\right)\sigma^2 + 2t_1t_2\left(\sum a_ib_i\right)\sigma^2\right\}\right.$$

which is a MGF of BN ($\left(\sum a_i\right)\mu, \left(\sum b_i\right)\mu, \left(\sum ai^2\right)\sigma^2, \left(\sum bi^2\right)\sigma^2, \rho\right)$

$$\rho = \frac{\sum a_ib_i\sigma^2}{\sigma^2\sqrt{\sum a_i^2}\sqrt{\sum b_i^2}} = \frac{a'b}{\|a\|\|b\|}$$

By uniqueness of MGF, $(U,V) \sim \text{BVN}\left((\Sigma a_i)\mu, (\Sigma b_i)\mu, h\left(\Sigma a_i^2\right)\sigma^2, \left(\Sigma b_i^2\right)\sigma^2, \frac{\sum a_ib_i\sigma^2}{\sigma^2\sqrt{\sum a_i^2}\sqrt{\sum b_i^2}}\right)$ with $\rho = \frac{a'b}{\|a\|\|b\|} = \cos\theta$, which θ is the angle between a and b. $(U,V) \sim BVN; U, V$ are independent distributed iff $\rho = 0$ iff,

$$\sum a_ib_i = 0$$
$$\Rightarrow a'b = 0$$
$$\Rightarrow a \text{ is } \perp \text{ to } b.$$

b) Here $\bar{X} = \frac{1}{n}\sum X_i = \sum a_iX_i, ai = \frac{1}{n}\forall i$

$$\text{and } X_k - \bar{X} = \sum b_iX_i \Rightarrow b_i = \begin{cases} 1 - \frac{1}{n}, & i = k \\ -\frac{1}{n}, & i \neq k \end{cases}$$

$X_k - \frac{1}{n}\sum X_1 = -\frac{1}{n}X_1 - \frac{1}{n}X_2 - \cdots + \left(1 - \frac{1}{n}\right)X_k - \frac{1}{n}X_{k+1}\cdots - \frac{1}{n}X_n$
are linear combination of $X_1, X_2, \ldots, X_n$ if $N\left(\mu, \sigma^2\right)$.

$$E(\bar{X}) = E\left(\frac{1}{n}\sum X_i\right) = \frac{1}{n}\sum E\left(X_i\right) = \frac{n\mu}{n} = \mu$$

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n}\sum X_i\right) = \frac{1}{n^2}\sum V\left(X_i\right) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

$$\text{cov}\left(X_k, \bar{X}\right) = \text{cov}\left(X_k, \frac{X_1 + \cdots + X_n}{n}\right) = \frac{\sigma^2}{n}$$

$$\text{Var}\left(X_k - \bar{X}\right) = \text{Var}\left(X_k\right) + \text{Var}(\bar{X}) - 2\,\text{cov}\left(X_k, \bar{X}\right) = \sigma^2 + \frac{\sigma^2}{n} - \frac{2\sigma^2}{n} = \left(1 - \frac{1}{n}\right)\sigma^2$$

$$\text{cov}\left(\bar{X}, X_k - \bar{X}\right) = \text{cov}\left(\bar{X}, X_k\right) - \text{var}(\bar{X}) = \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = 0$$

Hence, $\left(\bar{X}, \left(X_k - \bar{X}\right)\right) \sim \text{BVN}\left(\mu, 0, \frac{\sigma^2}{n}, \left(1 - \frac{1}{n}\right)\sigma^2, 0\right)$
Hence, $\bar{X}$ and $\left(X_k - \bar{X}\right)$ are independently distributed, $k = 1(1)n$.
$\Rightarrow \bar{X}$ and $\left(X_1 - \bar{X}, X_2 - \bar{X}, \ldots X_k - \bar{X}\right)$ are independently distributed.
So, $\bar{X}$ and $\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$ are independently distributed.
$\Rightarrow \bar{X}$ and $(n-1)s^2$ are independently distributed (Proved).